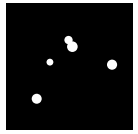
Module 2 - stage 2 fine-tuning step (train_module2_poc.py)

G initialised from stage 1 (final-G.ckpt);
D fresh (spectral norm)



X
baseline image
(3ch) + Module 1
mask (1ch)



concat, 4+c_dim ch

Generator



G(x)
synthesized
image, target
stage (3ch)

Discriminator

after every epoch, 10 % held-out
eyes / patients: L1(G(x), Reg(Y))
− 0.1 · Module 1 agreement
with c → lowest = best-G.ckpt
(100 epochs, early stop)

d_loss = fake − real + 10·GP
+ CE(D_cls(Y), c) [clinical c]
g_loss = adv + λ_rec·L1(G(x), Reg(Y))
+ CE(D_cls(G(x)), c)
λ_rec = 10, λ_adv = λ_cls = 1
n_critic 5, lr 1e-4, batch 8

c stage, one-hot
= FOLLOW-UP grade (target)

s = c (AdaIN style vector)



Y
real follow-
up (raw)

Affine registration



Reg(Y)
follow-up
warped to
X's geometry

L1
rec

pairs: FIRE + LongDR + Tianjin (progressing pairs oversampled to 30 %); registration trained once, self-supervised